

Lab Sheet 11: Real-Time Mobile Camera Integration via Python & OpenCV

Course: Computer Vision Lab

Objective: Stream live video from a mobile phone camera to a PC environment, process the incoming feed using Python and OpenCV, and apply real-time computer vision techniques.

Prerequisites & Initial Setup

To connect a mobile camera feed to Python, students can use IP streaming apps (such as IP Webcam on Android or iVCam / EpocCam on iOS/Android) over the same Wi-Fi network.

1. Live Motion & Security Alarm Detector

- **Task:** Track frame differences in real time. If motion above a specific pixel threshold is detected (e.g., someone walking past the phone camera), trigger a visual on-screen alert and save an emergency snapshot to disk.
- **Key Skills:** cv2.absdiff(), frame subtraction, contour area thresholding.

2. Mobile QR Code & Barcode Scanner for Inventory

- **Task:** Turn the smartphone into a portable handheld scanner. Read QR/barcodes in real time from the stream, decode the data, and log the decoded items into a CSV file with timestamps.
- **Key Skills:** OpenCV QRCodeDetector or pyzbar, CSV file manipulation.

3. Hand Gesture Desktop Controller

- **Task:** Point the mobile phone at the user's hand. Detect hand landmarks and translate finger count or gestures into system actions (e.g., controlling PC master volume, switching slides, or scrolling).
- **Key Skills:** Mediapipe Hands, gesture mapping, pyautogui.

4. Smart Attendance System via Face Recognition

- **Task:** Place the phone at an entryway. Detect faces in the stream, match them against a local directory of known student/employee faces, and mark attendance automatically.
- **Key Skills:** face_recognition library or OpenCV Haar Cascades, facial feature embeddings.

5. Document Scanner with Perspective Transform

- **Task:** Use the phone camera to photograph documents lying on a table. Automatically detect the four corners of the sheet, apply a perspective warp to flatten it, and convert it to a high-contrast black-and-white scan.
- **Key Skills:** Canny edge detection, cv2.findContours(), cv2.getPerspectiveTransform().

6. Portable Color & Defect Inspector

- **Task:** Point the smartphone camera at objects on a desk. Isolate specific color ranges in the HSV color space (e.g., detecting ripe vs. unripe fruit, or identifying defective parts on a mini assembly line).
- **Key Skills:** RGB to HSV conversion, color thresholding (cv2.inRange), masking.

7. Virtual Whiteboard / Air Painting App

- **Task:** Hold a colored object (like a bright blue marker cap or a laser pointer) in front of the phone camera. Track its position continuously and draw a path on the PC screen to paint or write in mid-air.
- **Key Skills:** Object tracking, drawing primitives (cv2.line, cv2.circle), canvas buffers.

8. Real-Time Optical Character Recognition (OCR) Text Reader

- **Task:** Aim the phone at printed text (e.g., book pages, street signs, product labels). Extract the text live and display the recognized string as a dynamic overlay on the video feed.
- **Key Skills:** Pytesseract (Tesseract OCR API), image thresholding, text overlay (cv2.putText).

9. Mobile Driver Drowsiness / Blink Monitor

- **Task:** Mount the phone facing a person's face. Track facial landmarks to calculate the Eye Aspect Ratio (EAR). If the user closes their eyes for longer than a specified duration, play an audible alarm through the PC speakers.
- **Key Skills:** Mediapipe Face Mesh or dlib, Euclidean distance calculation, audio triggers (winsound / pygame).

10. Real-Time Augmented Reality (AR) Filter Overlay

- **Task:** Detect facial features or specific planar surfaces using the phone feed, then render a custom 2D/3D visual overlay (e.g., virtual glasses, hats, or 3D bounding boxes) mapped to target anchors in real time.
- **Key Skills:** Landmark localization, image overlaying with alpha channels (transparency blending).